

# Integrated Sensing and Backscatter Communication: Toward Self-Sustainable Autonomous Internet of Things Networks

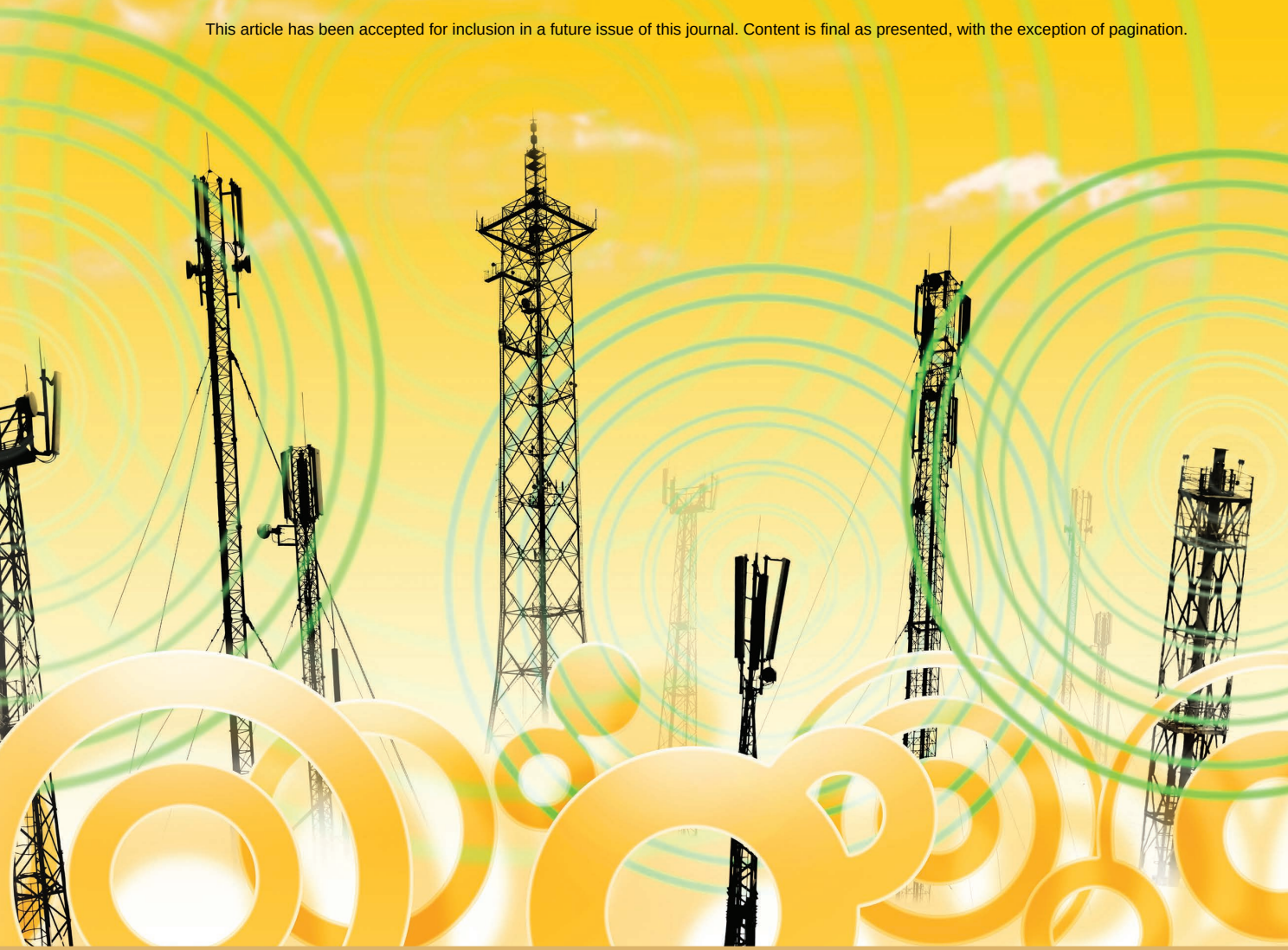
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**B**ackscatter communication has emerged as a promising enabler for battery-free and low-complexity Internet of Things (IoT) devices, and advances in radar sensing and millimeter-wave (mm-wave) systems have expanded its capabilities beyond identification toward localization, monitoring, and environmental awareness. More recently, these trends have converged into integrated sensing and backscatter communication (ISABC), where the same reflected waveform supports

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both data transmission and sensing. This article reviews the foundations and recent progress of ISABC, with emphasis on frequency-modulated continuous-wave (FMCW) radar-based implementations, which is meaningful to clarify how range estimation, tag identification, and data transfer can be jointly realized within a unified passive architecture. The article highlights both the opportunities and current limitations of ISABC and argues that future systems will evolve toward spatially aware, waveform-coordinated, and electromagnetically engineered backscatter platforms for battery-free sensing and 6G integrated wireless intelligence.

## Introduction

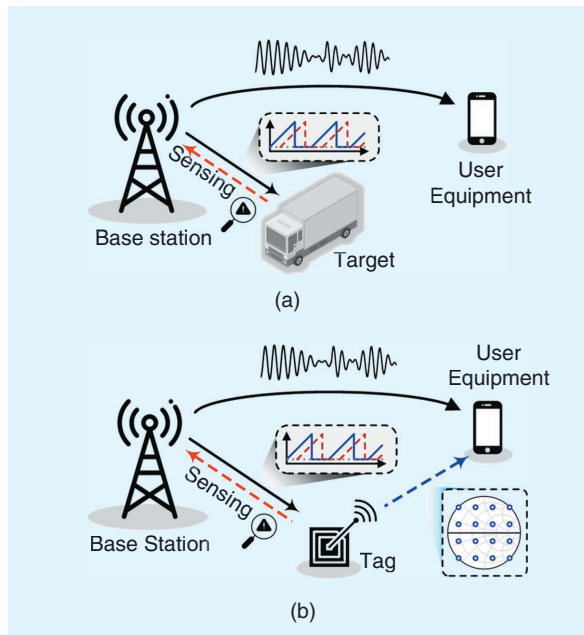
Over the past decades, backscatter communication has emerged as one of the most energy-efficient wireless communication technologies for IoT and wireless sensor networks [1], [2], [3]. Unlike conventional wireless communication systems that actively generate RF signals using oscillators and power amplifiers, backscatter communication enables passive devices to transmit information by modulating and reflecting incident

electromagnetic waves [2]. This mechanism significantly reduces power consumption and hardware complexity, making backscatter communication particularly attractive for battery-free sensors, massive IoT deployments, and low-cost wireless devices [1].

Early backscatter systems were primarily developed for RF identification (RFID) systems, where passive tags communicate with readers by modulating their radar cross section (RCS). Over time, backscatter technology has evolved beyond RFID into wireless sensing, localization, and environmental monitoring systems. Recent advances in mm-wave radar and passive backscatter tags have enabled high-resolution localization, motion tracking, and infrastructure sensing applications [4], [5], [6], [7], [8].

More recently, researchers have begun integrating sensing and communication functionalities into unified backscatter systems. This concept, often referred to as *ISABC*, enables simultaneous data communication and environmental sensing using the same backscatter hardware platform. Several works have demonstrated integrated radar backscatter communication systems





**Figure 1.** (a) Conventional integrated sensing and communications and (b) the developed ISABC.

using FMCW radar signals and passive backscatter tags [9], [10], [11], [12].

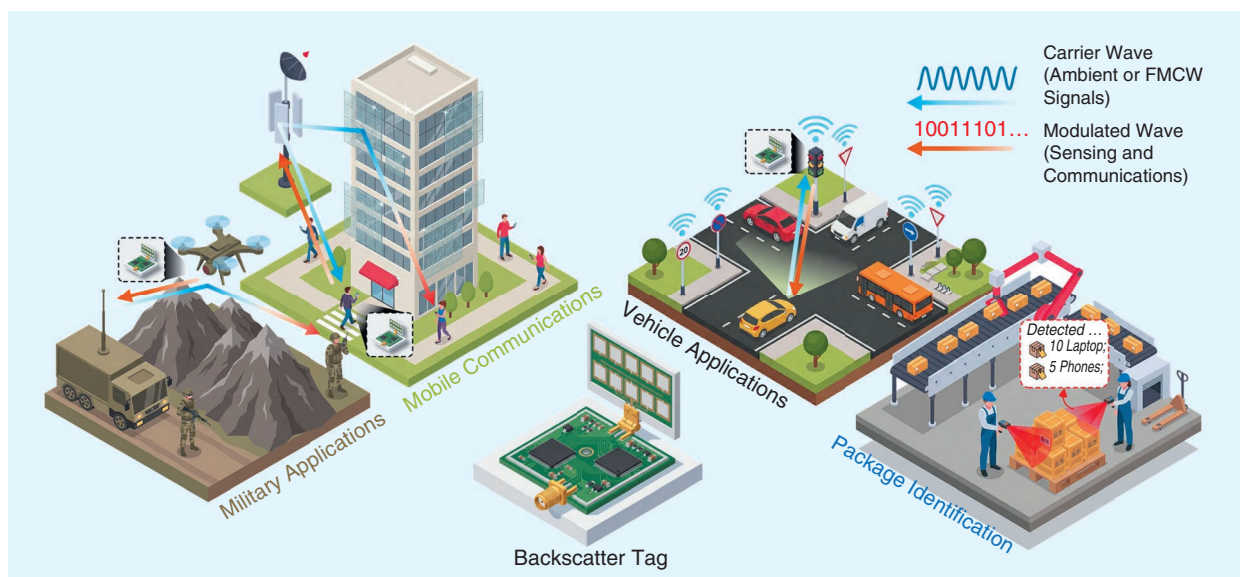
As shown in Figure 1(a), this concept is closely related to the recently emerging concept of integrated sensing and communication (ISAC), which is considered a key technology for future 6G wireless systems [10]. However, ISABC fundamentally differs from conventional ISAC systems [10]. In traditional ISAC systems, both sensing and communication are performed using active transceivers that generate RF signals and share waveform, spectrum, and hardware resources

[10]. In contrast, ISABC systems rely on passive backscatter tags that do not generate RF signals but instead modulate and reflect incident radar waveforms, as in Figure 1(b) [9]. As a result, ISABC systems enable ultralow-power ISAC, making them particularly suitable for battery-free IoT sensors, smart environments, and large-scale distributed sensing network, as demonstrated in Figure 2 [1], [13], [14].

The evolution of backscatter technology can be viewed in several stages: RFID-based identification systems, backscatter communication networks, backscatter sensing systems, and, finally, ISABC systems. This evolution reflects the increasing demand for low-power wireless communication and sensing technologies in IoT, smart city, and autonomous system applications [1], [3], [10].

Despite significant progress in backscatter communication, sensing backscatter systems and communication backscatter systems have historically been developed independently [3]. Sensing backscatter systems primarily focus on localization, tracking, and environmental sensing using radar signals and backscatter tags [4], [5], [6], [7], [8]. In contrast, communication backscatter systems focus on data transmission, modulation techniques, and network architectures for passive communication [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]. However, both types of systems rely on the same physical principle—modulation of reflected electromagnetic waves through impedance switching [2]. This observation naturally leads to the integration of sensing and communication into a unified backscatter system, which forms the basis of ISABC.

In recent years, several research efforts have demonstrated ISABC systems leveraging FMCW radar



**Figure 2.** ISABC systems provide ultralow-power ISAC for several applications proper for ultralow-power communication and large-scale distributed sensing networks.

waveforms, chirp-based signaling, and mm-wave platforms, enabling a unified architecture for simultaneous communication and sensing [9], [10], [11]. These systems exploit passive backscatter tags to jointly perform range, velocity, and angle estimation while embedding information modulation, as evidenced by recent works on high-precision localization and beam-aware backscatter sensing [4], [6], [7], [8]. Such integration opens new opportunities for smart environments, autonomous systems, and large-scale IoT deployments, where battery-free operation and minimal maintenance are critical requirements.

Despite recent advances, current ISABC systems still suffer from limited communication throughput, severe double path loss, limited spatial multiplexing capability, and coverage degradation in blockage environments. Furthermore, since sensing and communication often share the same waveform and hardware platform, a fundamental tradeoff exists between the sensing accuracy and communication data rate.

Future ISABC systems are therefore expected to evolve toward multiantenna backscatter systems, distributed sensing networks, metasurface-based backscatter platforms, and networked radar–communication architectures. Multiple-input, multiple-output (MIMO) backscatter systems can improve spectral efficiency and communication range via spatial multiplexing and beamforming [27], [28], while distributed multitag networks enable large-scale environmental sensing and cooperative localization [25], [29]. In addition, reconfigurable intelligent surfaces and metasurface-based backscatter structures can improve coverage in blockage environments and enhance link budgets by controlling electromagnetic wave propagation [30], [31], [32], [33]. These technologies are expected to constitute foundational enablers of sensing-integrated wireless architectures in future IoT and 6G networks, where joint spectrum utilization, hardware sharing, and passive operation are anticipated to become core system requirements.

The remainder of this article is organized as follows. The “Background: Backscatter and FMCW Radar Signal Models” section introduces the fundamentals of backscatter communication and signal models, including impedance modulation and backscatter link analysis. The “Sensing Backscatter and Communication Backscatter Systems” section reviews sensing backscatter systems and mm-wave backscatter localization techniques.

The “ISABC Systems” section discusses ISABC systems, including FMCW radar backscatter and joint sensing and communication architectures. The “Toward Robust High-Rate ISABC: Modulation, Spatial Expansion, and Reflective Hardware” section reviews backscatter modulators, modulation techniques, and MIMO backscatter systems in the context of ISAC. Finally, the “Conclusion” section concludes the article and discusses future research directions for ISABC systems.

## Background: Backscatter and FMCW Radar Signal Models

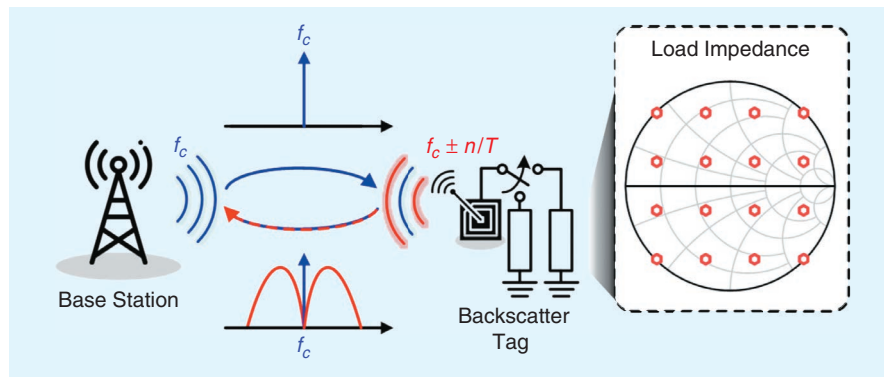
ISABC systems combine backscatter communication and radar sensing into a unified signal processing framework. To understand ISABC systems, it is necessary to review the signal models of backscatter communication and FMCW radar systems. This section introduces the signal model of backscatter system, the FMCW radar ranging principle, and a unified signal model that explains how sensing and communication can be performed simultaneously in ISABC systems.

### Backscatter Signal Model

As illustrated in Figure 3, a backscatter signal operates by modulating and reflecting an incident electromagnetic wave instead of generating a new RF signal, where the tag modulates an externally supplied carrier by changing its reflection coefficient [34]. The carrier signal from the reader radiates a nonmodulated continuous-wave (CW) at a carrier frequency at  $f_c$ . The backscatter tag connected to a backscatter modulator alters the tag impedance for the incident wave. The frequency of the modulated backscattered field is located in the sidebands of the backscattered field at  $f_c \pm (n/T)$ , where  $T$  represents the symbol period. This process is commonly referred to as *load modulation* [35].

The reflection coefficient of a tag antenna is defined as

$$\Gamma = \frac{Z_L - Z_A^*}{Z_L + Z_A} \quad (1)$$



**Figure 3.** Backscatter communication between a tag and a reader. The tag is designed to modulate its impedance among  $N$  different states.

where  $Z_L$  is the load impedance connected to the antenna and  $Z_A$  is the antenna impedance. By switching the load impedance among different values, the reflection coefficient becomes time varying and can be used to encode information.

Assuming that the reader transmits a CW signal, the transmitted signal can be expressed as

$$s(t) = A \cos(2\pi f_c t). \quad (2)$$

The signal propagates from the reader to the tag and then back to the reader after reflection. Therefore, backscatter communication experiences a two-way propagation channel. The received signal at the reader can be written as

$$r(t) = h_{rt} h_{tr} \Gamma(t) s(t) \quad (3)$$

where  $h_{rt}$  and  $h_{tr}$  represent the reader-to-tag and tag-to-reader channels, respectively. This equation shows that a backscatter signal suffers from double path loss due to the two-way propagation between the reader and the tag, which is one of the main limitations of backscatter system range.

### FMCW Radar Signal Model

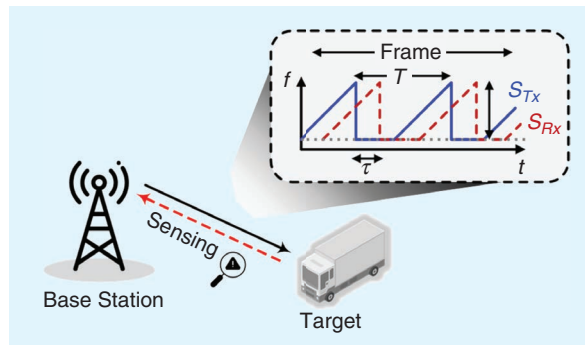
In ISABC systems, sensing is typically performed using FMCW radar. FMCW radar transmits a chirp signal whose frequency increases linearly over time. As represented in Figure 4, the range of a target can be estimated by measuring the frequency difference between the transmitted and received signals [10].

The transmitted FMCW chirp signal can be expressed as

$$s(t) = A \cos\left(2\pi\left(f_c t + \frac{K}{2} t^2\right)\right) \quad (4)$$

where  $f_c$  is the starting frequency and  $K$  is the chirp slope, defined as

$$K = \frac{B}{T} \quad (5)$$



**Figure 4.** The sensing scenario using FMCW radar.

where  $B$  is the frequency bandwidth and  $T$  is the chirp duration. If the signal is reflected from a target located at distance  $R$ , the round-trip delay is

$$\tau = \frac{2R}{c} \quad (6)$$

where  $c$  is the speed of light. After mixing the transmitted ( $S_{tx}$ ) and received ( $S_{rx}$ ) signals, the beat frequency is obtained as

$$f_b = K\tau = \frac{2RK}{c}. \quad (7)$$

Thus, the target range can be estimated from the beat frequency as

$$R = \frac{cf_b}{2K}. \quad (8)$$

This is the fundamental ranging principle of FMCW radar systems.

### ISABC Signal Model

In ISABC systems, the FMCW radar waveform transmitted from the reader is reflected by passive or semi-passive backscatter tags. The tag does not generate an RF carrier by itself; instead, it embeds communication data by changing its time-varying reflection coefficient, denoted by  $\Gamma(t)$ . Therefore, the received signal at the reader contains both radar sensing information and backscatter communication data.

The signal backscattered from a tag can be expressed as

$$r(t) = h_{rt} h_{tr} \Gamma(t) \cos\left(2\pi\left[f_c(t - \tau) + \frac{K}{2}(t - \tau)^2\right]\right) \quad (9)$$

where  $h_{tr}$  and  $h_{rt}$  denote the forward and backward propagation channels, respectively;  $\Gamma(t)$  is the time-varying reflection coefficient of the tag; and  $\tau = 2R/c$  is the round-trip delay for a tag located at range  $R$ .

At the reader, the received signal is mixed with a copy of the transmitted FMCW signal and then low-pass filtered. The high-frequency mixing product around  $2f_c$  is removed, and the remaining dechirped beat signal can be written as

$$r_b(t) = A\Gamma(t) \cos(2\pi f_b t + \phi_0) \quad (10)$$

where  $A$  includes the channel and scattering amplitudes,  $\phi_0$  is a constant phase term, and  $f_b$  is the FMCW range beat frequency.

Assuming that the tag has a radial velocity  $v$ , the reflected signal also experiences a Doppler frequency shift. The Doppler frequency is defined as

$$f_d = \frac{2v}{\lambda} = \frac{2vf_c}{c} \quad (11)$$

where  $\lambda = c/f_c$  is the wavelength. Under the slow-motion approximation within one chirp, the Doppler shift can be included in the dechirped signal as

$$r_b(t) = A\Gamma(t) \cos(2\pi(f_b + f_d)t + \phi_0). \quad (12)$$

Here, the sign of  $f_d$  depends on the radial motion direction, and a positive  $f_d$  is assumed for notational simplicity.

The communication data are embedded in the time variation of the tag reflection coefficient. For a modulation component at  $f_{m,k}$ , the reflection coefficient can be expressed as

$$\Gamma(t) = \Gamma_0 + \beta_k \cos(2\pi f_{m,k}t + \theta_k) \quad (13)$$

where  $\Gamma_0$  is the static reflection component,  $\beta_k$  is the modulation amplitude,  $f_{m,k}$  is the  $k$ th tag modulation frequency, and  $\theta_k$  is the corresponding modulation phase.

Substituting  $\Gamma(t)$  into the dechirped signal gives

$$\begin{aligned} r_b(t) = & A\Gamma_0 \cos(2\pi(f_b + f_d)t + \phi_0) \\ & + \frac{A\beta_k}{2} \cos(2\pi(f_b + f_d + f_{m,k})t + \phi_0 + \theta_k) \\ & + \frac{A\beta_k}{2} \cos(2\pi(f_b + f_d - f_{m,k})t + \phi_0 - \theta_k). \end{aligned} \quad (14)$$

This result shows that the tag modulation produces sideband components around the range-Doppler beat frequency  $(f_b + f_d)$ . Thus, the received frequency components are

$$f_{rx,k}^+ = f_b + f_d + f_{m,k} \quad (15)$$

$$f_{rx,k}^- = f_b + f_d - f_{m,k}. \quad (16)$$

If the upper modulation sideband is used for communication, the received communication-bearing component can be simply represented as

$$f_{rx,k} = f_b + f_d + f_{m,k}. \quad (17)$$

Here, the tag modulation shifts the range-Doppler beat component by  $f_{m,k}$ , enabling communication data recovery from the same dechirped baseband signal.

### Key Insights for ISABC Systems

From the unified signal model, ISABC systems reuse the radar waveform for communication while simultaneously performing sensing. The tag does not generate RF signals but modulates the incident waveform, enabling ultralow-power operation. Range information is obtained from the beat frequency, velocity

information from the Doppler frequency, and communication data from modulation frequencies. As a result, sensing and communication share the same waveform, spectrum, and hardware infrastructure.

This unified signal model forms the theoretical foundation of ISABC systems and explains how passive tags can simultaneously support localization, sensing, and data communication.

### Sensing Backscatter and Communication Backscatter Systems

Sensing backscatter systems and communication backscatter systems have historically been developed in parallel but largely independent research directions. Sensing backscatter systems primarily focus on localization, tracking, and environmental sensing using radar signals and passive tags, while communication backscatter systems focus on data transmission, modulation techniques, and network architectures for passive wireless communication. Despite these different research objectives, both systems rely on the same physical principle, namely, modulation of reflected electromagnetic waves through impedance switching. This observation naturally leads to ISABC systems, which are discussed in the next section. Before discussing integrated systems, this section reviews representative sensing backscatter systems and communication backscatter systems.

#### Sensing Backscatter Systems

Sensing backscatter systems use radar signals and passive backscatter tags to perform localization, tracking, and environmental sensing. In these systems, the reader transmits radar signals, such as FMCW chirps, and passive tags reflect and modulate the signal. The reader then processes the received signal using range fast Fourier transform (FFT), Doppler FFT, or angle estimation techniques to estimate the position and motion of the tags.

One of the representative sensing backscatter systems is Hawkeye, which demonstrates long-range localization using backscatter tags. Hawkeye achieves a median localization accuracy of approximately 7 mm at a range of 160 m by leveraging FMCW radar and backscatter tags that modulate the reflected signal with specific frequency signatures. The system distinguishes multiple tags by assigning different modulation frequencies to each tag, which appear as separate peaks in the frequency domain after FFT processing [5].

R-fiducial uses FMCW radar and backscatter tags to distinguish different tags using both the range FFT and Doppler FFT. Each tag modulates the reflected signal with a unique modulation pattern, and the radar system separates the tags by analyzing the frequency components in both the range and Doppler domains



[6]. Similarly, Millimetro uses the Doppler FFT to distinguish tags based on their modulation frequencies in the backscattered FMCW signal. The system demonstrates that multiple backscatter tags can be separated in the Doppler domain even when they are located at similar ranges [7]. Retroreflective radar-readable road signs (ROSSs) introduce polarization-based backscatter tags that can be detected reliably in roadway environments. Instead of relying solely on frequency modulation (FM), ROSSs use the geometric layout of the tag and polarization rotation in the reflection pattern to improve tag detection and robustness in practical environments [8].

The sensing range in Table 1 should be interpreted together with the channel condition and the tag/radar architecture. SuperSight focuses on indoor non-line-of-sight (NLOS) localization, where the signal reaches the tag through specular reflections rather than a direct LOS path. Therefore, its reported 8-m range is not a long-range detection target but a design tradeoff for achieving subcentimeter 3D localization accuracy under NLOS multipath conditions. In contrast, R-fiducial reports a shorter measured range of 25 m because it targets low-latency and wide-field-of-view (FOV) traffic infrastructure detection rather than narrow-beam long-range localization; however, its range can be extended through chirp combination and a higher effective radar link budget. Millimetro achieves long-range localization up to 100 m by assuming a mainly LOS radar-to-tag path and by using a Van Atta array-based retroreflective tag, which concentrates the incident radar energy back toward the reader. Overall, the different ranges mainly reflect the tradeoff among the effective isotropic radiated power (EIRP), tag RCS/

retroreflective gain, antenna FOV, coherent integration time, and LOS/NLOS channel conditions.

### Communication Backscatter Systems

In contrast to sensing backscatter systems, communication backscatter systems focus on wireless data transmission using passive backscatter tags. The main research objectives in communication backscatter include increasing the data rate, improving the communication range, designing efficient modulation schemes, and developing network architectures for passive communication systems.

One of the early communication backscatter systems was the backscatter wireless sensor network with wireless power transfer (WPT), which proposed a system that combines backscatter communication with WPT using two different frequencies. As detailed in Figure 5, one frequency is used for energy harvesting, while the other frequency is used for backscatter communication, enabling continuous operation of passive sensor nodes without batteries [36], [37], [38], [39]. A key feature of this work is that the rectifier and backscatter paths are codesigned so that the node can continue harvesting energy even during communication, thereby avoiding the severe duty cycle limitation of conventional passive tags. This architecture established an important foundation for self-sustained backscatter nodes in which sensing, powering, and uplink transmission can be supported simultaneously.

As examined in Figure 6(a), mmTag demonstrated that mm-wave backscatter links can support high-throughput and low-power communication through passive beam alignment and polarization-based self-interference suppression, thereby providing an

**TABLE 1. Sensing backscatter systems.**

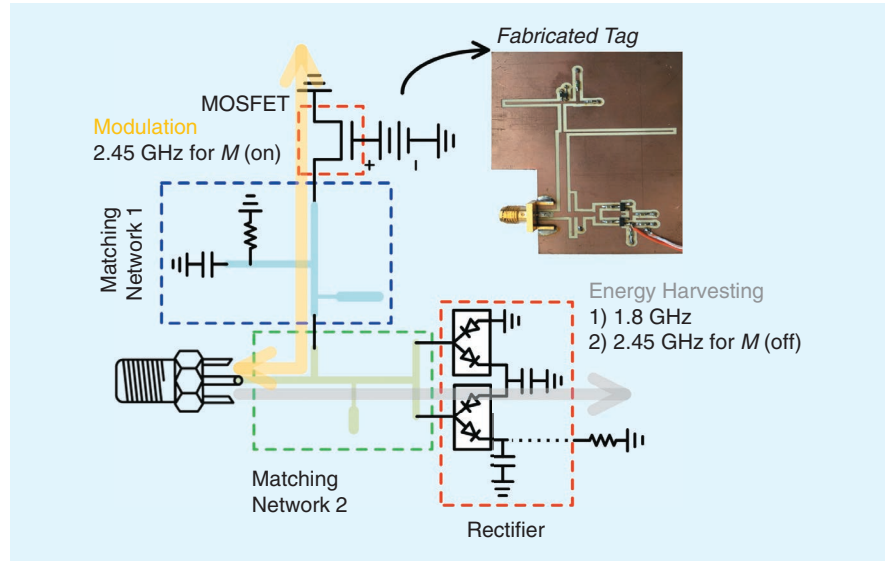
| Work           | Frequency/<br>Bandwidth                                                 | Range                                                                                        | Sensing/Localization<br>Performance                                                                    | Supported<br>Function |
|----------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------|
| SuperSight [4] | 77-GHz FMCW radar, 4-GHz bandwidth                                      | 8 m, measured in indoor NLOS scenarios with impenetrable blockages and specular reflectors   | Subcentimeter 3D NLOS localization, position accuracy up to 5.7, 5.5, and 7.7 mm in x, y, and z at 8 m | NLOS localization     |
| R-fiducial [6] | 24–24.3-GHz FMCW radar                                                  | 25 m, measured for traffic infrastructure detection; 100% detection up to 25 m with 120° FOV | Median range error of 0.1 m and median angle error of 6°                                               | Sensing/localization  |
| Millimetro [7] | 24-GHz FMCW radar, 250-MHz bandwidth; conceptually extendable to 77 GHz | 100 m, measured in LOS indoor/outdoor scenarios; >90% detection rate up to 100 m             | Median localization accuracy of 15 cm over 100 m, median range error of 10–20 cm up to 100 m           | Localization          |

NLOS: non-line of sight; FOV: field of view.

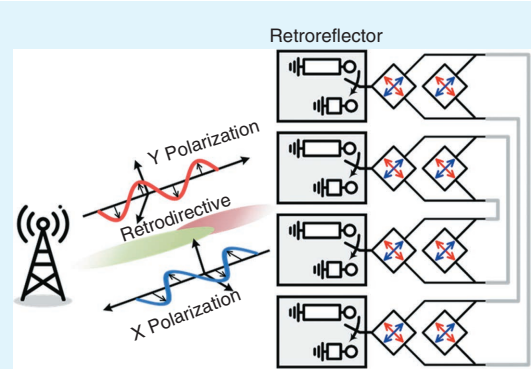
important hardware and propagation foundation for later mm-wave backscatter communication systems [16]. In this system, the backscatter node modulates and reflects the received signal while also changing the polarization and reflecting the signal back toward the direction of arrival, improving the link efficiency and communication range. More specifically, mmTag adopts a Van Atta-type retro-directive structure so that the reflected beam automatically returns toward the access point without requiring active phased arrays or beam training at the tag. In addition, polarization conversion allows the access point to separate the weak backscatter signal from the strong transmitted signal, which is essential for practical full-duplex-like operation at mm-wave frequencies. The performance trend in Figure 6(b) further illustrates how the achievable data rate scales with the link distance and received power, highlighting the practical tradeoff between throughput and range in mm-wave backscatter systems.

Hardware-efficient orthogonal frequency-division multiplexing (OFDM) backscatter communication systems have also been proposed, where all-digital architectures are used to implement OFDM backscatter modulators using digital CMOS RF switches [19]. These systems demonstrate that high-order modulation schemes and multicarrier communication can be implemented using backscatter modulation without requiring active RF transmitters. Their main contribution is to show that digitally controlled discrete impedances can approximate single-sideband OFDM modulation, enabling a multipath-robust and spectrally flexible uplink with very low modulator power. This line of work is particularly important because it extends backscatter from simple single-carrier on-off keying (OOK)/frequency-shift keying (FSK) links toward configurable multicarrier physical layers that are compatible with modern wireless networking concepts.

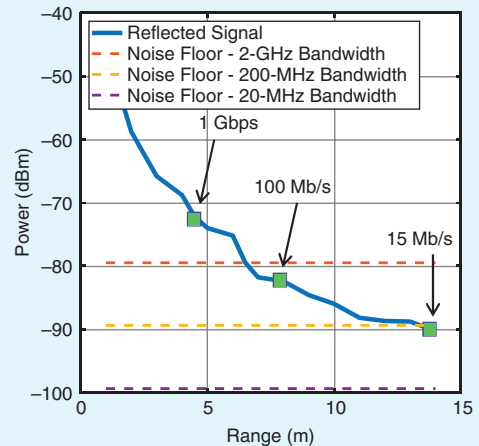
As reviewed in Figure 7, other communication backscatter research focuses on higher-order modulation schemes, such as quadrature amplitude modulation (QAM) backscatter modulation, multilevel impedance modulation, and frequency-shift backscatter modulation to improve spectral efficiency and the data rate [17], [18], [23]. These systems treat the passive tag as a communication transmitter that encodes digital information



**Figure 5.** Self-sustainable backscatter modulation with the combination of a dual-band rectifier and a conventional backscatter modulator [36].



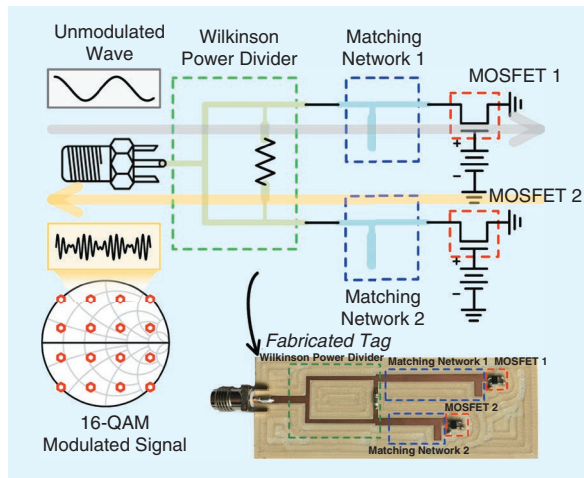
(a)



(b)

**Figure 6.** A mm-wave backscatter communication platform with passive beam alignment and polarization-based self-interference suppression: the (a) system model and (b) measured communication performance under different ranges of mmTag [16].





**Figure 7.** A scheme for 16-quadrature amplitude modulation (QAM) backscatter modulation using a Wilkinson power divider, which is effective to minimize the number of the transistors [17].

onto the reflected signal. For example, QAM backscatter modulators engineer multiple reflection coefficient states so that both the amplitude and phase can be controlled, enabling more than 1 b/symbol and, therefore, much higher spectral efficiency than binary switching. Similarly, in-phase and quadrature (I/Q) impedance modulation and chirp- or frequency-based backscatter schemes show that passive nodes can synthesize more advanced waveforms, such as long range (LoRA)-like chirps or multilevel ambient FM symbols, broadening the design space of passive communication beyond conventional RFID-style modulation.

Table 2 summarizes communication-oriented backscatter systems from the viewpoint of the carrier frequency, range, data rate, and modulation method. The comparison shows that high-data-rate backscatter is mainly achieved either by using wide mm-wave bandwidth or by employing high-order modulation in a controlled link budget. For example, mmTag and the printed mm-wave modulator demonstrate gigabit-per-second-class communication at short-to-moderate ranges by using directional mm-wave links and amplitude-shift keying (ASK)/QAM-type modulation [16], [21]. In contrast, low-frequency or ambient source systems, such as FM Rider and four-level pulse amplitude modulation (PAM) ambient FM backscatter, sacrifice the data rate to obtain a longer range or infrastructure-free operation by reusing broadcast FM signals [22], [23]. Other works, including OFDM backscatter and LoRa-compatible I/Q impedance modulation, focus on improving robustness or spectral efficiency, rather than maximizing only the raw data rate [18], [19], [27], [28]. Overall, communication-oriented backscatter systems exhibit a clear tradeoff among the data rate, operating range, modulation complexity, excitation power, and channel robustness.

## Comparison Between Sensing Backscatter and Communication Backscatter

Although sensing backscatter and communication backscatter systems have been developed independently, they share the same underlying physical principle: both systems modulate the reflection coefficient of a passive tag to encode information onto the reflected electromagnetic wave. The main difference lies in how the reflected signal is processed at the receiver.

In sensing backscatter systems, the receiver processes the signal using radar signal processing techniques, such as the range FFT, the Doppler FFT, and angle estimation, to extract localization and motion information. In communication backscatter systems, the receiver demodulates the signal to extract digital communication data.

In summary, sensing backscatter and communication backscatter systems are not fundamentally distinct but, rather, represent different signal processing strategies applied to the same reflected waveform. This observation directly motivates ISABC, which unifies both functions within a single passive backscatter architecture. This relationship is summarized in Table 3.

## ISABC Systems

### Concept of ISABC

ISABC systems aim to simultaneously perform localization, sensing, and communication using passive backscatter devices and radar waveforms. Unlike conventional backscatter communication systems, which focus only on data transmission, and sensing backscatter systems, which focus only on localization or environmental sensing, ISABC systems combine both functionalities into a unified signal processing framework.

The key idea of ISABC is that the backscattered signal contains multiple frequency components corresponding to range, velocity, and communication data. When an FMCW radar waveform is transmitted, the reflected signal from a passive backscatter tag produces a beat-frequency component that is related to the target range. If the tag modulates its reflection coefficient, additional frequency components appear around the beat frequency. These modulation sidebands can be used to encode communication data, while the beat frequency still provides range information.

Therefore, sensing and communication can be performed simultaneously by analyzing the frequency components of the received signal. Range information is obtained from the beat frequency, velocity information from the Doppler frequency, and communication data from modulation frequencies. This unified signal

**TABLE 2. Communication backscatter systems and modulators.**

| Work                                                     | Frequency/<br>Bandwidth                                     | Range/Test Scenario                                                          | Data Rate                                                                                | Modulation/EIRP or<br>Excitation Condition                                                                                                                                               |
|----------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mmTag [16]                                               | 24-GHz ISM band,<br>250-MHz available<br>bandwidth          | 4.6-, 8-, and<br>14-m measured<br>communication range                        | 1 Gb/s at 4.6 m, 100<br>Mb/s at 8 m, 10–15 Mb/s<br>at 14 m                               | Amplitude-shift keying<br>backscatter, access point<br>transmit power of 12 dBm,<br>including antenna gain                                                                               |
| Ultrafast<br>backscatter<br>modulator [20]               | Backscatter: 2.45 GHz;<br>WPT: 1.7 GHz                      | Modulator-level<br>characterization,<br>wireless link range N/R              | 960 Mb/s, with 8.37%<br>EVM; 61.5 fJ/b                                                   | 16-QAM, EIRP N/R, WPT input<br>power of 5 dBm at 1.7 GHz                                                                                                                                 |
| Printed mm-wave<br>modulator [21]                        | 24–28 GHz                                                   | 0.5-m laboratory<br>backscatter link                                         | 1-Gb/s BPSK, 1-Gb/s<br>QPSK, 2-Gb/s 16-QAM;<br>0.17 pJ/b                                 | BPSK, QPSK, and 16-QAM; EIRP<br>N/R; 24.5-GHz CW illumination<br>with high-gain horn antennas                                                                                            |
| Bipolar CMOS<br>mm-wave<br>backscatter<br>modulator [26] | 20–28 GHz                                                   | On-wafer/modulator<br>characterization,<br>wireless link range N/R           | Static EVM characterization<br>for 16/32/64-QAM,<br>multigigabit-class link<br>potential | 16-QAM, 32-QAM, and 64-<br>QAM; EIRP N/R; input power<br>characterized at 0, –10, and<br>–20 dBm                                                                                         |
| OFDM<br>backscatter<br>modulator [19]                    | 2.45-GHz ISM band                                           | 1-m bistatic over-the-<br>air validation                                     | 1.25 Mb/s using five BPSK<br>subcarriers, 195 kb/s over-<br>the-air validation           | Five-subcarrier BPSK OFDM<br>backscatter, EIRP N/R, 0-dBm<br>CW source with RF amplification<br>and 10-dBi antenna                                                                       |
| I/Q impedance<br>LoRa backscatter<br>[18]                | 2.45 GHz, LoRa-<br>like CSS, 125-kHz<br>available bandwidth | 10-m LOS, 7.5-m<br>NLOS, and 10-m<br>wall-obstructed indoor<br>scenarios     | Raw LoRa-like rate of<br>approximately 6.8 kb/s                                          | LoRa-compatible CSS via I/Q<br>impedance modulation, EIRP<br>N/R, required carrier input<br>power of –40 dBm (LOS), –25<br>dBm (NLOS), and –26 dBm<br>(wall) for error-free transmission |
| FM Rider [22]                                            | Ambient FM band,<br>88–108 MHz                              | 58 m for maximum-<br>bit-rate case, 107-m<br>maximum tag-to-<br>reader range | 333 b/s at 58 m, with BER<br>of 2.85%; 200 b/s at 107<br>m, with BER of 6%               | In-band binary FSK ambient FM<br>backscatter, EIRP N/R, ambient<br>FM station 34.5 km away, with<br>measured received FM power<br>around –40 dBm                                         |
| Four-PAM<br>ambient FM<br>backscatter [23]               | Ambient FM<br>broadcasting band,<br>bandwidth N/R           | 1-m tag-to-reader<br>indoor proof of<br>concept, FM station<br>34.5 km away  | 345 b/s, 78.2 nJ/b at 345<br>b/s                                                         | Four-level PAM ambient FM<br>backscatter, EIRP N/R, ambient<br>FM carrier used instead of<br>dedicated RF source                                                                         |

N/R: not reported; ISM: Industry, Science, Medicine band; BPSK: binary phase-shift keying; QPSK: quadrature phase-shift keying; CSS: chirp spread spectrum; BER: bit error rate; PAM: pulse amplitude modulation; EVM: error vector magnitude.

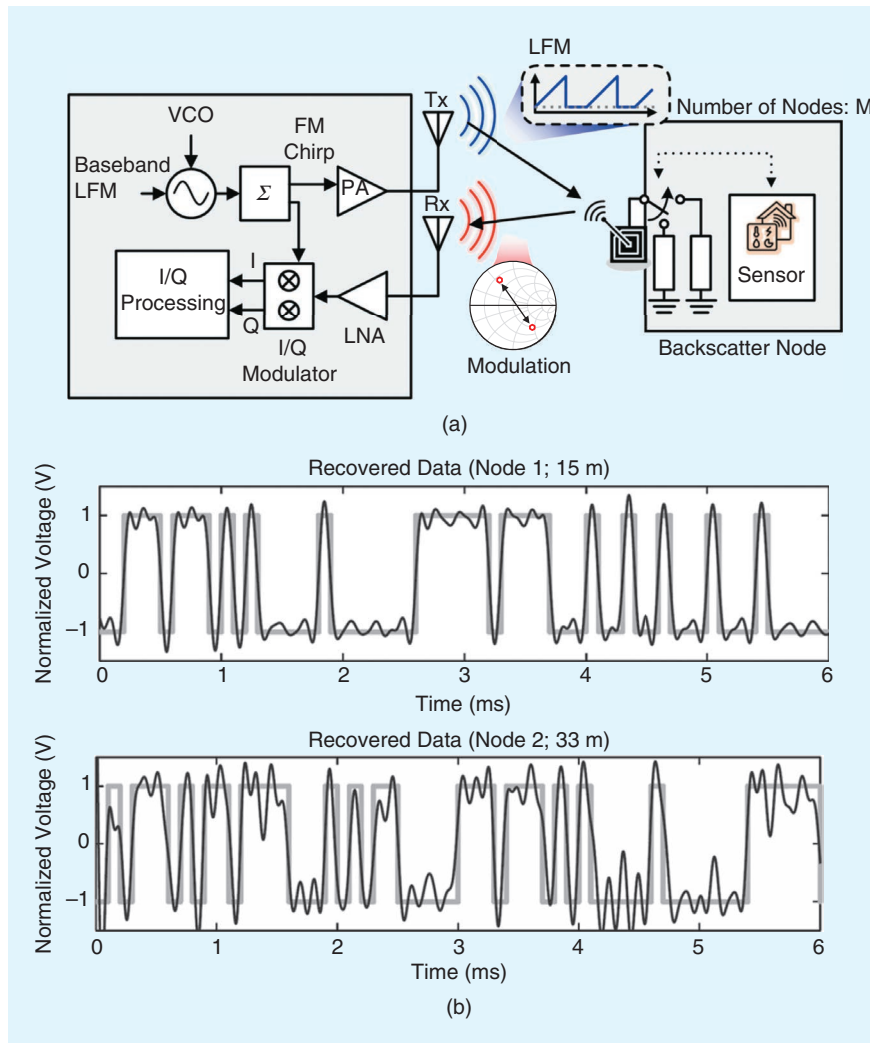
**TABLE 3. A classification of backscatter systems according to functionality and signal processing.**

| System Type                  | Objective                                                | Signal Processing                                             |
|------------------------------|----------------------------------------------------------|---------------------------------------------------------------|
| Sensing backscatter          | Localization, tracking, and environmental sensing        | Range FFT, Doppler FFT, and angle estimation                  |
| Communication<br>backscatter | Data transmission between tag and reader/receiver        | Demodulation and decoding for encoded FMCW<br>signal from tag |
| ISABC                        | Simultaneous localization, sensing, and<br>communication | FFT-based sensing pipeline combined with<br>demodulation      |

processing framework enables passive backscatter devices to support localization, sensing, and communication simultaneously without active RF transmitters.

Another important characteristic of ISABC systems is that the same hardware platform can be used for both sensing and communication. The radar

transmitter, receiver, antennas, and signal processing blocks are shared between sensing and communication functions. This reduces system complexity, power consumption, and hardware cost, making ISABC systems suitable for large-scale IoT sensing networks and future 6G ISAC systems.



**Figure 8.** A multichannel FMCW-based ISABC system: (a) the system model and (b) a comparison between the recovered and transmitted data waveforms for two nodes at different ranges [10]. VCO: voltage-controlled oscillator; LFM: linear FM; PA: power amplifier; LNA: low-noise amplifier; Tx: transmitter; Rx: receiver.

### ISABC Systems Using FMCW Radar

Several representative studies have shown that an FMCW radar can be extended from a pure sensing platform into an ISABC system by allowing passive or semi-passive tags to modulate the reflected radar waveform. The key principle is that the FMCW signal already provides range information through the beat frequency after dechirping, while the tag embeds communication data by altering the spectral content of the reflected waveform through impedance switching, frequency shifting, or waveform-aware modulation. As a result, a single interrogating radar waveform can simultaneously reveal the tag position and recover its data stream.

One of the earliest representative examples is the multichannel FMCW backscatter system in [10]. In this work, the radar transmits standard FMCW chirps, and each tag modulates the incident signal through a simple load

switching backscatter mechanism, as in Figure 8(a). The sensing function is obtained from the conventional FMCW beat-frequency principle: after mixing the received echo with the transmitted chirp, the radar extracts a beat tone whose frequency is proportional to the round-trip delay and, hence, to the tag range. Communication is superimposed on top of this process because the tag intentionally introduces modulation-induced spectral components on the reflected FMCW waveform. By separating these components in the frequency domain, the radar can decode the uplink data while preserving range estimation. The recovered waveforms in Figure 8(b) further confirm that reliable data demodulation is maintained over different ranges, demonstrating the robustness of simultaneous sensing and communication in practical scenarios. An important contribution of this work is that multiple tags can be separated by assigning different modulation frequencies, so the same FMCW radar can simultaneously perform ranging and multichannel telemetry from distributed passive sensor nodes.

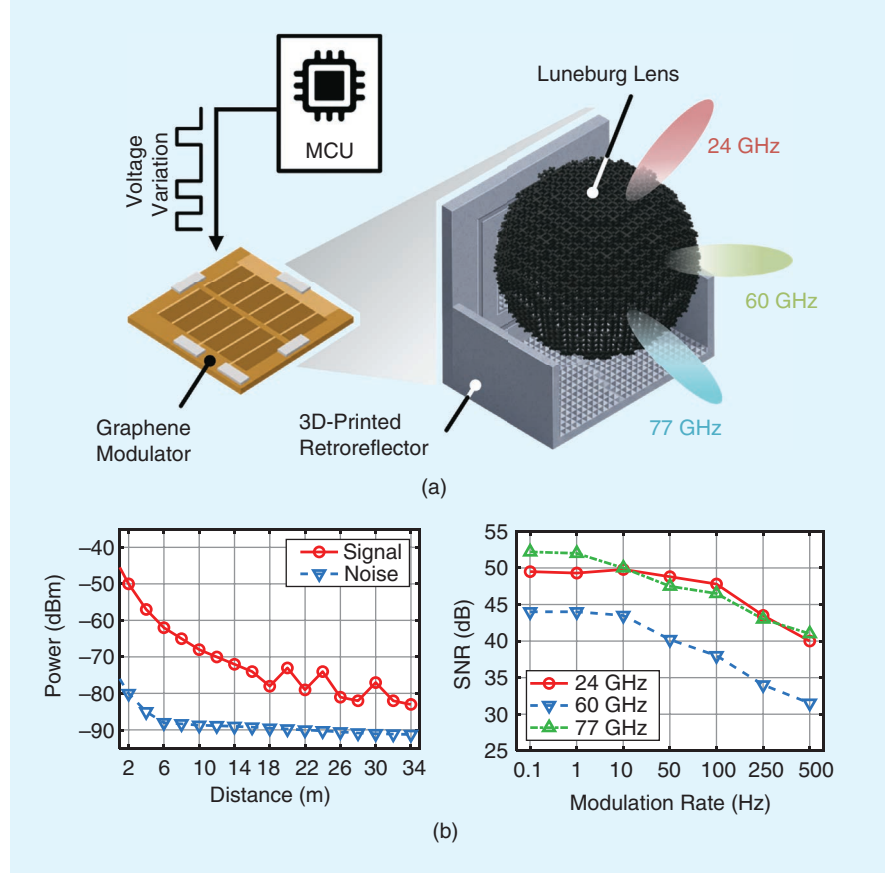
A more advanced mm-wave realization is UniScatter in [11], which addresses a different but equally important limitation: the hardware scalability of radar backscatter tags across a wide bandwidth and wide angular coverage. UniScatter consists of two tightly coupled hardware components, namely, a graphene-based modulator and a Luneburg lens-based retroreflector, as in Figure 9(a). The communication function is achieved by changing the reflection coefficient of the graphene metasurface so that the reflected radar signal carries modulation sidebands or coded spectral signatures. The sensing function remains radar-centric: an FMCW radar estimates the range from the beat-frequency spectrum while also identifying the tag through the imposed modulation pattern. The main innovation, however, lies in the tag architecture itself. Unlike conventional resonant antenna tags that



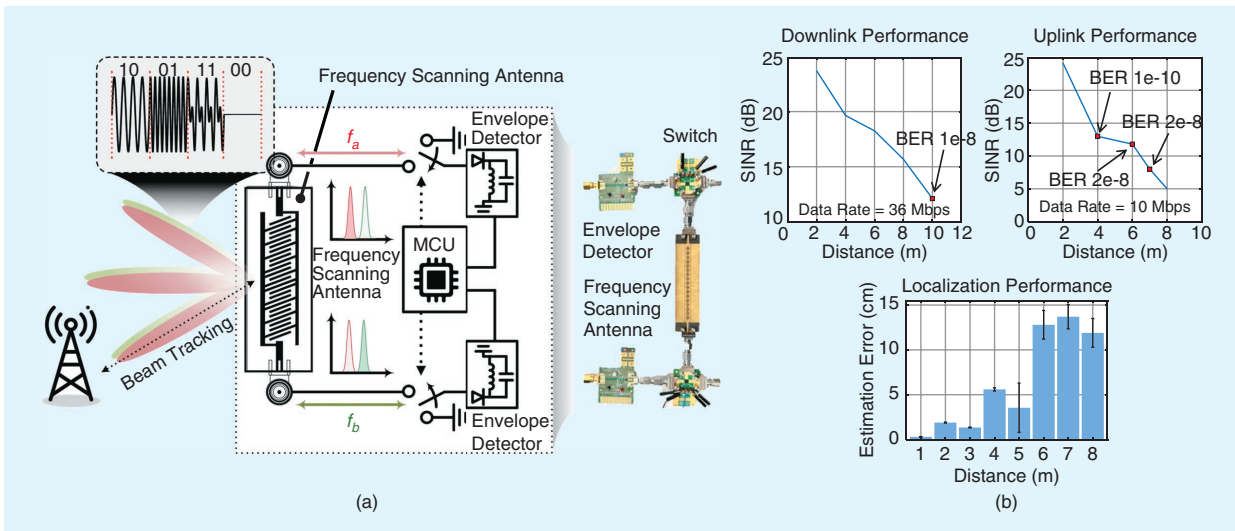
work only over a narrow band and limited angular region, UniScatter maintains strong and consistent backscatter from 24 to 77 GHz and over a wide 3D FOV. The Luneburg lens improves the retroreflection and RCS, while the graphene modulator preserves the controllable reflection over a broad frequency range. The measurement results in Figure 9(b) further validate that the tag maintains a clear signal-to-noise margin and stable performance across distance, frequency, and the modulation rate, confirming its broadband and angle-robust operation. Therefore, UniScatter shows that ISABC performance is not determined only by the waveform and digital signal processing but also by how the tag hardware is engineered to support broadband angle-robust joint sensing and communication.

Another important milestone is MilBack, which expands the scope of integrated backscatter systems from uplink-only telemetry to uplink, downlink, and localization in the same mm-wave platform [15]. As depicted

in Figure 10(a), MilBack introduces a passive node architecture that can operate in both absorptive and



**Figure 9.** An advanced backscatter tag based on a metamaterial lens and graphene modulator for a mm-wave ISABC system: (a) UniScatter structure backscattering FMCW signals within the multifrequency and (b) the localization and communication performance of UniScatter [11]. MCU: microcontroller unit.

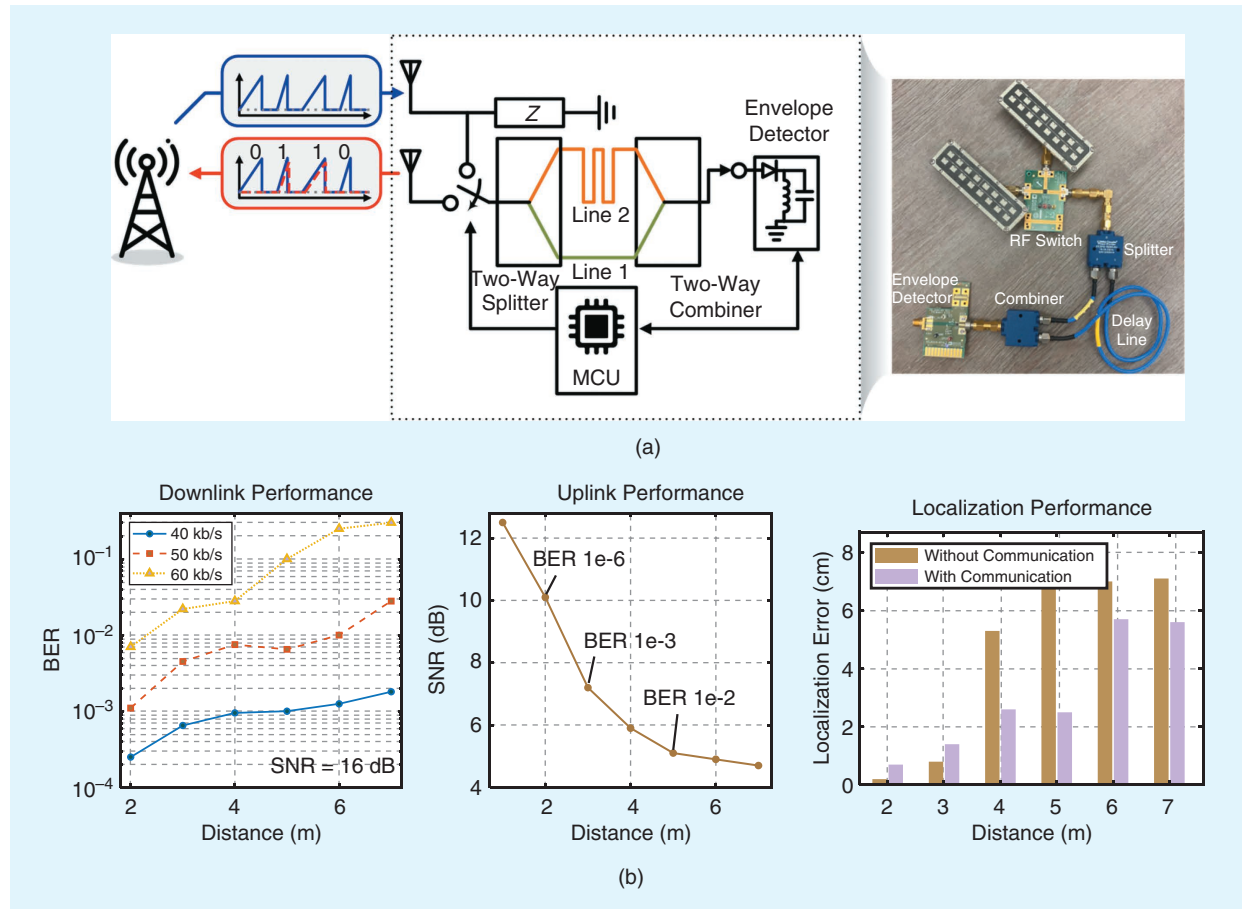


**Figure 10.** A mm-wave ISABC system for two-way communication and localization: the (a) system model, with an inset showing the node implementation, and (b) localization and communication performance of MilBack [15]. SINR: signal-to-interference-plus-noise ratio; SNR: signal-to-noise ratio; BER: bit error rate.

reflective modes. In the reflective mode, the node backscatters the incident mm-wave signal toward the access point, enabling uplink communication. In the absorptive mode, the node receives energy from the interrogating signal and extracts downlink information. Localization is enabled by exploiting the mm-wave signal structure and the directional behavior of the tag so that the access point can estimate position and orientation from the reflected spectral response. For communication, MilBack proposes a low-complexity modulation strategy that avoids conventional I/Q demodulation overhead and instead uses a frequency-based symbol formation that is more suitable for passive mm-wave nodes, as described in Figure 10(a). The corresponding performance evaluation in Figure 10(b) further demonstrates the achievable communication reliability and localization accuracy as a function of distance, highlighting the practical feasibility of the proposed integrated architecture. The approach's significance lies in showing that the backscatter tag can be more than a simple reflective endpoint: by codesigning the node structure, beam behavior, and

modulation method, the same hardware can support localization together with two-way communication.

The most complete realization of this idea is BiScatter in [9], which explicitly targets simultaneous uplink communication, downlink communication, radar sensing, and localization using off-the-shelf FMCW radars. As shown in Figure 11, the key downlink innovation is chirp slope-shift keying (CSSK), where information is embedded into the radar waveform itself by varying the slope of the FMCW chirp. A passive tag equipped with low-power differential circuitry and delay lines observes the resulting beat-frequency change and decodes the downlink message without requiring an active RF receiver, as in Figure 11. At the same time, the tag performs uplink communication by modulating its reflected signal, so the radar can recover the tag's data from the backscatter spectrum while still performing conventional FMCW range estimation. In this sense, sensing and communication are not merely overlaid but are jointly designed through the waveform, tag circuit, and radar receiver processing chain. BiScatter is particularly important because it demonstrates that



**Figure 11.** A mm-wave ISABC system based on chirp slope-shift keying and the low-power downlink decoding concept: the (a) system model, with an inset showing the node implementation, and (b) localization and communication performance of BiScatter [9].

a commercial FMCW radar can be transformed into a full ISABC node without abandoning radar compatibility, achieving centimeter-level localization and low-bit-error-rate two-way communication.

Communication can also be integrated into radar operation by modifying the radar waveform parameters themselves, rather than changing only the reflection coefficient of the tag. This idea is exemplified by the BiScatter framework, where the downlink is carried by chirp slope variation and the tag decodes it through passive delay-based frequency interpretation. This is an important conceptual step because it expands the ISABC design space beyond conventional impedance switching backscatter. In earlier systems, the tag primarily acts as a modulated reflector, whereas in this type of architecture, the radar waveform itself becomes a communication-bearing signal for the tag. As a result, the overall system supports a richer interaction model in which the radar not only senses the tag and receives its uplink response but also configures or controls the tag through embedded downlink signaling.

Taken together, these studies reveal a clear technical progression. The early FMCW backscatter work in [10] established the basic principle that beat-frequency processing can support simultaneous ranging and uplink telemetry. UniScatter then showed that practical mm-wave ISABC requires a new tag hardware paradigm with wideband modulation and robust retroreflection [11]. MilBack extended the architecture toward localization-aware and bidirectional backscatter

networking [15]. Finally, BiScatter unified radar sensing, localization, the uplink, and the downlink within a single FMCW-compatible framework [9]. Thus, the evolution of ISABC can be interpreted as a progression from range-plus-uplink sensing tags to broadband mm-wave reflective platforms and, finally, to fully interactive radar backscatter nodes that are jointly designed across waveform, hardware, and signal processing layers.

Table 4 further reports this progression from a system-level performance perspective. The multichannel FMCW backscatter system demonstrates the baseline feasibility of simultaneous ranging and uplink communication, but its data rate remains limited to the kilobit-per-second level because the design prioritizes compatibility with conventional FMCW radar processing [10]. UniScatter extends this concept to a much wider mm-wave frequency range from 24 to 77 GHz, showing that broadband and angle-robust tag hardware is essential for practical radar backscatter deployment, although its communication rate is still low because the emphasis is on robust tag design based on a metalelens and low-powered modulation across ultra-wide bandwidth [11]. MilBack shifts the design target toward a networked mm-wave backscatter node by supporting uplink, downlink, and localization with megabit-per-second-class communication at several-meter ranges [15]. BiScatter, in contrast, sacrifices the data rate for stronger FMCW radar compatibility and low-power passive downlink decoding while still

**TABLE 4. Representative ISABC systems.**

| Work                               | Frequency/Bandwidth                                                                 | Range-Dependent Data Rate                                                                                        | Sensing/Localization Performance                                                                                      | Supported Function                               |
|------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Multichannel FMCW backscatter [10] | 2.45 GHz, 40-MHz bandwidth                                                          | 10-kb/s sensor at 15 and 33 m                                                                                    | Simultaneous ranging and telemetry uplink for two sensor nodes in a cluttered indoor environment                      | Sensing and uplink communication                 |
| UniScatter [11]                    | Wideband response from 24 to 77 GHz, demonstrated with 24-, 60-, and 77-GHz devices | 20-b/s tag modulation, around 30-m communication distance, with signal-to-noise ratio > 10 dB and raw BER < 5.7% | Wide angular coverage over 70° in 3D, compatible with heterogeneous radar and joint communication and sensing devices | Sensing and uplink communication                 |
| MilBack [15]                       | 26.5–29.5 GHz, 3-GHz bandwidth                                                      | Downlink up to 36 Mb/s, uplink up to 10 Mb/s at 8 m and 40 Mb/s at 6 m, maximum uplink rate of 160 Mb/s          | Distance error below 5 cm at 5 m and below 12 cm at 8 m, median/90th-percentile angle error of 1.1/2.5°               | Localization and two-way communication           |
| BiScatter [9]                      | 9 GHz, with 1-GHz bandwidth; 24 GHz, with 250-MHz bandwidth                         | Two-way communication with BER < 10 <sup>−3</sup> up to 7 m, current data rate of about 50–100 kb/s              | Centimeter-level tag localization accuracy up to 7 m using off-the-shelf FMCW radars                                  | Sensing, localization, and two-way communication |

achieving centimeter-level localization and reliable two-way communication up to 7 m [9]. Therefore, the comparison in Table 4 shows that ISABC systems are not optimized along a single axis; instead, their performance is determined by the joint tradeoff among the radar compatibility, communication directionality, data rate, localization accuracy, operating range, and tag hardware complexity.

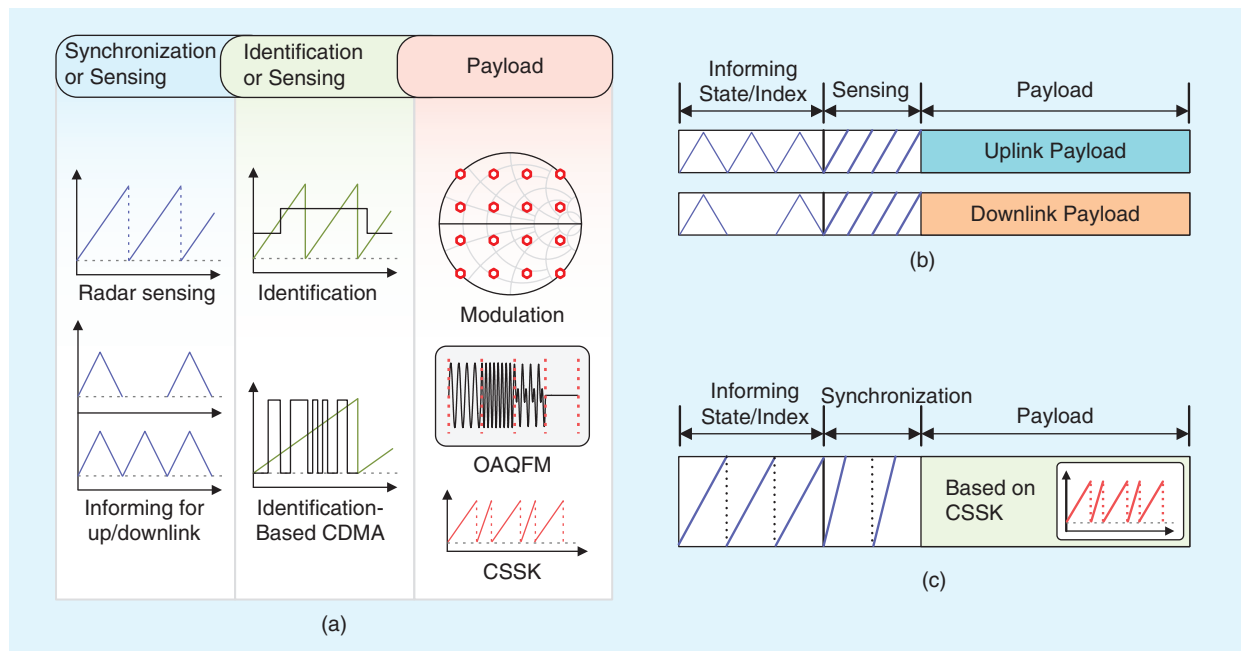
### Design Insights for ISABC Systems

The surveyed literature suggests that the evolution from sensing backscatter to ISABC is not merely a progression in modulation complexity but a transition in how the reflected radar waveform is functionally organized. In early sensing backscatter systems, the passive tag primarily acted as a radar-visible reflector with an embedded signature for identification. In more advanced ISABC systems, the same reflected waveform is further exploited to carry telemetry, uplink data, and, in some cases, even downlink interaction. From this perspective, as shown in Figure 12(a), the most important design insight is that modern ISABC can be interpreted as a structured coexistence of three functions within a single radar backscatter interaction: detection/synchronization, tag identification, and payload communication.

This interpretation becomes clear when comparing representative sensing backscatter systems with ISABC-oriented systems. In sensing-oriented designs, the tag modulation is rarely used for high-rate

communication; instead, it is mainly introduced to make the tag distinguishable from clutter and from other tags. For example, the multichannel FMCW backscatter system in [10] uses a low-rate preamble before the sensor data to improve range estimation and distinguish the node response from stationary clutter. It also employs range-dependent subcarriers for data recovery. Therefore, although the system is not described using the terminology of a structured radar backscatter frame, it already includes the essential elements. Likewise, R-fiducial assigns each tag a unique spreading code and jointly estimates identity and distance through correlation over range bins, effectively treating the reflected waveform as an identification-bearing radar response rather than a pure communication signal [6]. Millimetro similarly separates multiple tags by assigning different modulation frequencies so that the switching signatures appear as distinguishable components in the range–Doppler profile [7]. In all these systems, the modulation is not only a sensing aid but also an implicit label attached to the radar target.

In contrast, ISABC systems extend this role of modulation from identification to information transfer. The basic FMCW backscatter architecture in [10] demonstrated that the dechirped signal can simultaneously preserve range information and recover telemetry from the backscatter spectrum. Later systems strengthened this principle in different directions. UniScatter showed that robust ISABC operation depends not only on waveform processing



**Figure 12.** The unified functional packet structure for complex ISABC situations involving two-way communication and the existence of multitag: the (a) description of the general packet structure, (b) timing-aware packet structure [9], and (c) link-state-informing packet structure [15]. CDMA: code-division multiple access; OAQFM: orientation-assisted quadrature FM.



but also on tag hardware that sustains controllable reflection over a broad frequency range and wide angular coverage, using a graphene-based modulator and a Luneburg lens retroreflector for wideband radar backscatter interaction [11]. More broadly, the ISABC concept formalized in [12] clarified that the passive tag is no longer treated as a mere radar target but as a sensing target and information source simultaneously. This changes the role of the reflected waveform: it must support radar observables, such as range, velocity, or angle, while also carrying information intended for a user or infrastructure node [12].

Taken together, these studies suggest that an effective ISABC architecture should not be understood as simple “data modulation on top of FMCW” but, rather, as a layered waveform interaction in which different portions or features of the reflected signal serve different system functions. A useful abstraction is to interpret the backscatter response in a frame-oriented manner: a known or structured component supports detection and synchronization, a dedicated signature-bearing component supports tag identification and multitag separation, and a payload-bearing component conveys sensor data or communication symbols. Importantly, this does not necessarily require explicit packetization in the conventional digital communications sense. Instead, it means that the radar backscatter system should be designed so that these three functions are separable in processing, whether through time segmentation, modulation frequency allocation, code-domain separation, chirp-to-chirp patterning, or waveform parameter embedding.

This viewpoint leads to several concrete design implications. First, identification should be treated as a first-class design objective rather than a byproduct of communication. The sensing backscatter literature shows that reliable tag identification is fundamental for clutter suppression, multitag operation, and low-latency detection, especially in radar-centric environments where many passive reflectors coexist [6], [7]. Second, as indicated in Figure 12(b), communication modulation must be designed with awareness of radar observables, such as link states (uplink/downlink) and synchronization [9], [15]. Since the dechirped FMCW signal already maps propagation delay into beat frequency, additional modulation should be placed such that it remains distinguishable from range- and Doppler-related components, rather than corrupting them [10]. Third, the hardware of the tag must be codesigned with this signal structure, as in Figure 12(c). As UniScatter illustrates, if the reflection coefficient, RCS, or angular response is unstable across frequency and direction, then neither identification nor data recovery can remain robust in practical deployments [11]. Finally, the most scalable future ISABC systems will likely be those that explicitly organize sensing,

identification, and communication as separate but coordinated layers of the reflected waveform, instead of assuming that a single undifferentiated modulation mechanism can serve all functions equally well.

Therefore, one of the key lessons from the surveyed works is that the path from backscatter sensing to full ISABC is not simply a shift from localization to communication but a progression toward functionally structured radar backscatter interactions. Early systems established how tags can be made visible and distinguishable to radar. ISABC systems then showed how those distinguishable reflections can also transport data. The next logical step is to systematically design radar backscatter waveforms and tag responses so that synchronization, identification, and communication are jointly supported within a unified framework. This perspective provides a clearer architectural basis for future ISABC systems intended for smart infrastructure, industrial sensing, autonomous mobility, and 6G sensing-aware IoT networks.

### **Toward Robust High-Rate ISABC: Modulation, Spatial Expansion, and Reflective Hardware**

The representative ISABC systems discussed so far establish the basic feasibility of using a single interrogating waveform for both radar sensing and backscatter communication. However, most existing demonstrations remain constrained by a limited payload rate, a weak backscatter link budget, narrow angular coverage, or poor multitag scalability. From this perspective, the next stage of ISABC development is not merely to combine sensing and communication in the same platform but to improve how efficiently information is embedded into the reflected waveform, how robustly it survives propagation, and how flexibly the tag hardware can shape the reflection itself [10], [11], [12].

A useful interpretation of the recent literature is that future ISABC performance will be determined by three tightly coupled design axes: waveform-efficient modulation, spatial expansion through MIMO and diversity, and reflection engineering through advanced modulator and metasurface hardware. These are not isolated research directions. Rather, they jointly determine whether ISABC can evolve from proof-of-concept radar tags into scalable platforms for dense sensing networks, smart infrastructure, and 6G sensing-aware IoT networks [2], [11], [12].

### **From Simple Impedance Switching to Waveform-Efficient Modulation**

A primary bottleneck of early backscatter communication was that the tag modulation mechanism was intentionally simple. Binary load switching based on OOK, ASK, binary phase-shift keying (BPSK), or low-order FSK minimizes circuit complexity and power

consumption, but it also restricts spectral efficiency and limits the amount of information that can be carried by the reflected waveform. This limitation becomes more critical in ISABC, where the same interrogation waveform must preserve radar observables, such as range and Doppler, while also carrying communication symbols [1], [2], [17]. As a result, improving the communication throughput of ISABC requires not only higher-order modulation at the tag but also a receiver and frame structure that remain compatible with radar processing.

This is why the backscatter literature gradually moved toward higher-order modulation. One representative example is the quadrature amplitude backscatter modulator of [17], which uses a Wilkinson power divider, a phase-shifted branch, and transistor-based impedance control to realize  $M$ -ary QAM signaling with passive backscatter hardware. In parallel, multilevel amplitude modulation schemes, such as four-level PAM, were introduced to improve spectral efficiency in ambient backscatter by creating multiple amplitude states in the reflected signal [23]. Although these studies were not originally developed for radar-integrated systems, they are directly relevant to ISABC because they demonstrate that the reflected waveform need not be limited to binary tag signatures. Instead, the backscatter response can be engineered as a communication-bearing signal with controllable amplitude and phase states [17], [23], [25]. In this sense, quadrature phase-shift keying (QPSK)- or QAM-based ISABC can be viewed as a natural extension of recent progress in waveform-efficient backscatter design.

At the same time, integrating QPSK or 16-QAM into an FMCW-based ISABC platform is not a straightforward extension of conventional backscatter modulation. After dechirp processing, the communication symbols appear on top of range-dependent beat components, rather than as an isolated baseband waveform. Therefore, a practical receiver must jointly perform range estimation, beat tone or subcarrier isolation, range-dependent subcarrier wipe off, and carrier/phase recovery before reliable QPSK or QAM demodulation becomes possible. This is fundamentally more demanding than binary detection because residual ranging error, clutter leakage, and phase distortion directly degrade the recovered constellation.

For this reason, waveform-efficient ISABC also requires a more structured frame design. A practical packet should include a radar-friendly preamble for detection and coarse synchronization, a tag ID or header field for node association, a pilot block for postdechirp phase and amplitude calibration, and a payload block carrying the higher-order symbols. Such a framing strategy is particularly important in distributed sensing scenarios, where multiple tags may occupy different ranges or partially overlapping

returns. The preamble supports robust detection in the radar domain, while the header and pilot enable coherent symbol recovery in the communication domain.

Another important transition is the move from carrier-oriented modulation to waveform-aware modulation. Since many ISABC platforms rely on FMCW radar, modulation schemes that are compatible with chirp-like incident signals are more attractive than amplitude-only methods designed for simple CW excitation. In this regard, chirp spread spectrum backscatter and I/Q impedance modulation are especially important. Belo et al. showed that an I/Q impedance modulator can generate LoRa-like linear FM chirps by properly varying the phase of the reflected wave, rather than directly synthesizing frequency at the tag [18]. Their implementation uses two RF transistors and quadrature combination so that the reflection phase spans a full  $360^\circ$  on an approximately constant-voltage-standing-wave-ratio circle, making chirp-compatible backscatter modulation feasible with low-power hardware [18]. Similarly, OOK chirp spread spectrum backscatter demonstrates that chirp-compatible backscatter can also be realized in WPT-assisted passive platforms without relying on a conventional active transmitter chain [40]. From an ISABC perspective, the importance of such approaches is not only that they can improve the data rate but also that they provide a natural framework for embedding communication symbols into radar waveforms that are already frequency structured [18], [40].

Taken together, these developments suggest that the next step for ISABC is not simply to replace BPSK with QPSK or 16-QAM but to codesign the tag modulator, radar waveform, packet structure, and receiver processing as a unified system. Only through such joint design can higher-order backscatter modulation increase the data rate without undermining the ranging and sensing functions that distinguish ISABC from conventional backscatter communication.

### ***Full-Duplex Modulator Hardware as the Enabler of Radar-Compatible Communication***

A radar-compatible ISABC link differs fundamentally from a conventional one-way backscatter uplink. The access node must continuously emit a sensing waveform, receive the weak tag-modulated echo, and preserve radar observables, such as range, Doppler, and angle. Therefore, the ISABC link should not be treated merely as a tag-to-reader telemetry channel. Instead, it requires a hardware architecture that can support simultaneous illumination, reception, and modulation within the same time-frequency resource. In this sense, full-duplex operation becomes an enabling condition for ISABC, rather than merely a protocol-level option achieved by adding handshaking procedures

or separating the uplink and downlink through time-division duplexing [9], [12], [15].

The need for full-duplex capability originates from the intrinsic carrier interference problem of in-band backscatter [41]. Since the desired backscatter response is a modulated version of the incident carrier, the receiver must detect an extremely weak reflected signal in the presence of a much stronger leakage or carrier component. This problem becomes more severe in mm-wave ISABC, where directional beams are used to compensate for path loss but transmitter-to-receiver leakage can still limit receiver gain and saturate the front end. Consequently, a radar or access point interacting with a backscatter tag must either employ full-duplex cancellation hardware or introduce an equivalent mechanism that separates the desired tag response from the strong self-interference [16].

Early full-duplex backscatter work showed that instantaneous feedback can be realized with simple passive analog components, enabling collision termination, rate adaptation, and bit-level retransmission without using power-hungry cancellation circuits [42]. Network-level extensions, such as time hopping spread spectrum, further demonstrated that full-duplex information transfer and wireless energy transfer can coexist in dense BackCom networks [43]. However, these approaches mainly address link-layer feedback or multiple-access interference. The techniques cannot fully remove the physical-layer carrier cancellation bottleneck required by radar-compatible backscatter, especially when the modulated waveform must remain available for sensing and the response of the backscatter tag must be recovered without sacrificing radar continuity.

Recent hardware-oriented studies therefore provide a more relevant direction for ISABC. Lotus revisits polarization mismatch and demonstrates that carrier interference can be suppressed in the analog domain through a backscatter-specific antenna and tag design, avoiding the spectrum overhead of frequency shifting and the power/protocol overhead of conventional cancellation approaches [44]. Although Lotus is not a complete full-duplex transceiver in the conventional active radio sense, it addresses the key physical bottleneck of in-band backscatter reception by exploiting polarization-domain separation between the direct carrier leakage and the desired tag-reflected echo. This enables carrier suppression without frequency shifting, thereby preserving spectrum efficiency while retaining the weak backscatter response.

Armstrong, reported in [45], takes the next step toward a truly ISAC-ready full-duplex backscatter platform at mm-wave frequencies. It introduces a full-duplex mm-wave backscatter tag architecture based on regenerative uplink amplification and regenerative

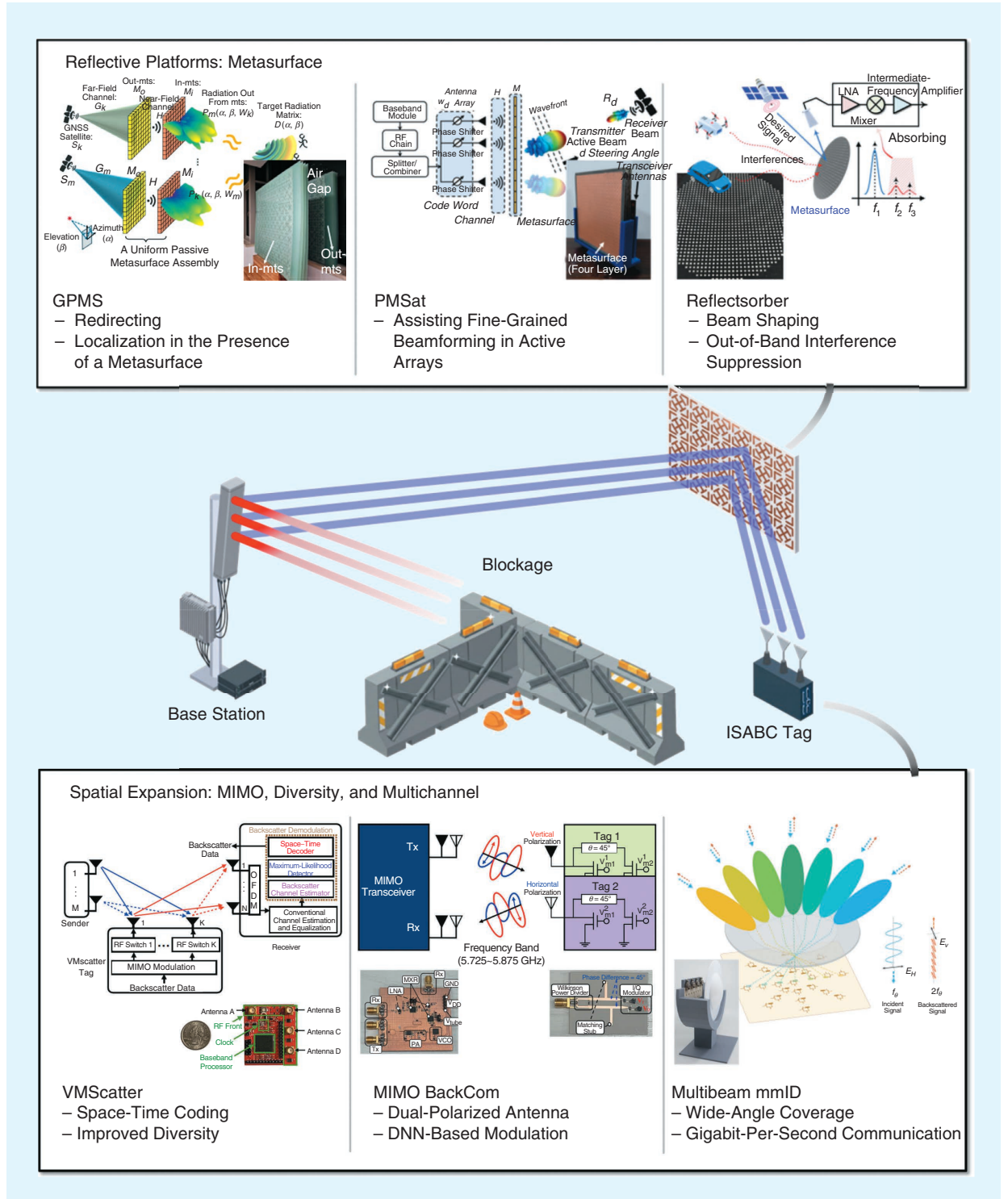
downlink reception, enabling simultaneous bidirectional communication at extended ranges with low-cost hardware [45]. This approach can particularly suggest the potential of full-duplex ISABC, where future tags should not only reflect a radar waveform for sensing but also receive modulated data while concurrently embedding backscattered signals into the reflected signal.

Taken together, these works suggest that the next stage of ISABC should move beyond half-duplex tags based on scheduling or purely protocol-level separation. Radar-compatible backscatter requires a full-duplex-capable hardware layer that can suppress carrier/self-interference, maintain continuous sensing illumination, and support bidirectional tag interaction. From this perspective, Lotus and Armstrong represent two complementary hardware paths: Lotus provides a low-overhead carrier cancellation principle for backscatter reception, while Armstrong demonstrates a practical mm-wave full-duplex tag design [44], [45]. Such full-duplex modulator hardware is therefore a key enabler for robust, high-rate, and sensing-compatible ISABC systems.

### ***Spatial Expansion in ISABC: MIMO, Diversity, and Angular Multiplexing***

If modulation determines how efficiently information is embedded into the reflected waveform, spatial expansion determines how reliably that information survives propagation and how cleanly it can be separated from other radar returns. As illustrated in the inset at the bottom of Figure 13, spatial degrees of freedom are essential not only for improving communication robustness and diversity through multiantenna backscatter techniques [27] but also for enabling polarization-domain multiplexing and dual-channel operation [28] as well as supporting angular selectivity and beam alignment for multitag separability [46]. From this perspective, MIMO is only one part of a broader design space that also includes polarization diversity, multibeam coverage, and angularly selective backscatter links.

On the communication side, backscatter systems have already shown that spatial processing can be exploited in multiple forms. MIMO backscatter architectures demonstrate that multiple reflecting elements and tag-side space-time coding can improve throughput and reduce the error probability by creating diversity-enhanced reflection paths. Versatile MIMO backscatter is particularly relevant because it shows that a backscatter system can approach the diversity gain of conventional MIMO even though the reflected symbols are not orthogonal in the usual active radio sense [27]. Likewise, the dual-channel polarization diversity backscatter platform in [28] shows that polarization can serve as an additional spatial degree of



**Figure 13.** Research directions for improving ISABC, based on prior backscatter and wireless communication studies. Top: reflective metasurface platforms for mitigating blockage and path loss [30], [31], [32]. Bottom: spatial expansion through MIMO, diversity, and multibeam techniques for improving the data rate [27], [28], [46]. Representative prior works are addressed in each panel. GNSS: global navigation satellite system; GPMS: GNSS positioning metasurfaces based system; DNN: deep neural network.

freedom for high-speed passive links. Together, these results suggest that future ISABC systems can use spatial diversity not only to increase the data rate but also

to improve the robustness of cascaded passive links that are inherently vulnerable to fading and weak backscatter returns [27], [28].



Recent lens-assisted mm-wave backscatter platforms further suggest that spatial expansion can contribute to higher throughput by enabling wide-angle multibeam operation in addition to conventional diversity gain. For example, lens-integrated mmID systems have demonstrated that broad angular coverage can be maintained together with high-order modulation and multigigabit backscatter transmission, while related fully passive and harvesting-oriented platforms indicate that such spatially extended operation can remain feasible even under passive or wireless-powered constraints [46], [47]. In parallel, sector-based beam alignment backscatter has shown that spatially partitioned coverage can also be exploited for adaptive directional links, where different sectors are selectively aligned for passive communication while sustaining 16-ary modulation and up to 387-kb/s transmission over 20 m [29]. Although these works are not yet formulated as FMCW-based ISABC systems, they imply that future ISABC may increase the data rate not only through higher-order modulation but also through the use of multiple beams, wider angular sectors, and richer spatial channels.

At the same time, bringing spatial expansion into FMCW-based ISABC is not a direct extension of conventional MIMO or beam-steered communication systems. In radar-integrated backscatter, each reflected component must be resolved jointly in range, Doppler, and angle after dechirp processing, rather than being treated as an isolated baseband spatial channel. This difficulty becomes even greater when multibeam or angularly multiplexed operation is introduced because multiple reflected paths may coexist within the same observation interval and can overlap in the radar cube if their delay, Doppler, or angular signatures are not sufficiently separable. As a result, spatially expanded ISABC requires the joint design of array processing, tag modulation orthogonality, chirp-to-chirp coherence, beam-domain separability, and radar cube association. Put differently, the benefit of multiple antennas, polarizations, channels, or beams is realized only when the reflected signals remain distinguishable in both the sensing domain and the communication domain. For this reason, the most promising role of spatial expansion in ISABC may be broader than spatial multiplexing alone: it can provide diversity gain for fragile passive links, angular resolution for localization, beam-level coverage extension, and multitag separability for distributed sensing, all within a common FMCW processing framework [27], [28], [29], [46], [47].

### ***From Passive Tags to Reflective Platforms: Metasurfaces and Reflection Engineering***

Even with higher-order modulation and spatial diversity, ISABC systems still face a fundamental bottleneck: the reflected signal is often too weak, too angle

sensitive, or too unstable across frequency to sustain robust sensing and communication in realistic environments. This shifts attention from signal processing alone to reflection engineering itself [3]. As illustrated in the top inset of Figure 13, metasurface-based reflective platforms provide a concrete realization of this concept by enabling programmable control over wavefront redirection, beam shaping, and frequency-selective response.

More recent metasurface studies suggest that the role of reflection engineering is not limited to strengthening the tag response but can be extended to creating or recovering useful propagation paths when the direct path is blocked, misaligned, or severely attenuated. At a broader system level, passive metasurface platforms further indicate that engineered reflective structures can actively create usable propagation paths, rather than merely enhance local reradiation. PMSat shows that a passive metasurface, when jointly optimized with a small phased array, can realize fine-grained beam steering in the uplink and wide-angle focusing in the downlink, improving the signal-to-noise ratio by 8.32 and 16.57 dB, respectively [30]. Likewise, GNSS positioning metasurfaces based system (GPMS) demonstrates that passive metasurfaces can redirect weak global navigation satellite system signals from outdoors into indoor spaces and scatter them to enlarge coverage, thereby increasing the number of visible satellites and significantly improving indoor localization accuracy [31].

From an ISABC viewpoint, these results are important because they suggest that metasurfaces can be used to secure reflective paths in blockage-limited or shadowed environments, effectively extending the observable region of the radar backscatter link, rather than simply strengthening an already available LOS reflection. This reflective freedom, however, comes with a new challenge for localization. In FMCW-based ISABC, an intentionally redirected or multihop reflection may appear as an additional propagation path with a different delay and angle, which can distort the range-angle association if it is interpreted as a direct target return. In other words, the same metasurface that improves communication coverage may also create geometric ambiguity in sensing. Related mm-wave backscatter localization studies already suggest both sides of this tradeoff: retroreflective structures can substantially improve detectability and range performance, but specular or redirected paths must be explicitly modeled rather than treated as nuisance-free echoes [4], [7], [11].

This perspective is also consistent with recent integrated metasurface concepts for wireless systems, where reflective surfaces are expected not only to steer useful signals but also to suppress or reshape undesired electromagnetic components through

frequency-selective reflection and absorption [32]. For this reason, metasurface-assisted ISABC should be developed together with reflection-aware sensing algorithms. Future systems will likely need path labeling or environment-assisted inference to distinguish direct returns from engineered reflections, and they may have to estimate not only the tag state but also the reflection state of the propagation channel. One possible direction is to treat the reflective surface as a known controllable node in the sensing loop so that the radar does not interpret redirected energy as an unknown target but as a structured auxiliary path. In this view, metasurfaces are not merely hardware extensions for stronger backscatter. They become channel shaping components whose benefits for blockage mitigation, coverage extension, and signal enhancement must be balanced against the need for reliable localization and interpretable radar observables [30], [31], [32].

### **A Unified Perspective for Future ISABC Systems**

Taken together, these developments point to a broader conclusion: future ISABC systems will likely advance not through a single dominant innovation but through the joint evolution of modulation, spatial processing, and reflective hardware. Higher-order and waveform-aware modulation increases how much information can be embedded into a radar-compatible reflection [17], [18], [23], [40]. Advanced modulator circuits determine whether such modulation is physically realizable under passive or low-power constraints [21], [26], [36]. MIMO and polarization diversity techniques enlarge the spatial degrees of freedom available for both sensing and communication [27], [28]. Metasurfaces and intelligent reflective tags further improve the link budget and reduce fading sensitivity by engineering the reflection process itself [11], [33].

Accordingly, the future of ISABC can be interpreted as a progression from simple modulated radar reflectors toward high-capacity reflective platforms. In early systems, the tag primarily served as a distinguishable radar target. In more advanced systems, it became a telemetry source. In the next generation, it is likely to become a programmable sensing and communication surface whose waveform response, spatial behavior, and reflection characteristics are jointly optimized. Such a progression is essential if ISABC is to support dense multitag sensing networks, robust operation in blocked or fading-prone environments, and higher-rate interaction with passive devices in future 6G and smart infrastructure scenarios [9], [11], [12], [27].

### **Conclusion**

ISABC is emerging as a compelling extension of backscatter technology from simple identification links to multifunctional passive wireless platforms. By allowing

the same reflected waveform to carry both sensing observables and communication data, ISABC offers a unique path toward battery-free and low-complexity wireless nodes that can support localization, environmental awareness, and information transfer within a common hardware framework. Recent demonstrations based on FMCW radar, multichannel ranging, mm-wave retroreflective tags, and wideband reflective structures show that this concept is no longer only theoretical but is steadily moving toward practical system realization.

At the same time, the field is still in an early stage. Fundamental challenges remain in overcoming double path loss, maintaining reliable communication without corrupting radar observables, separating multiple tags in shared spatial and spectral domains, and preserving performance under blockage, angular misalignment, and hardware constraints. These limitations suggest that future progress will depend not only on improved modulation or signal processing but also on deeper codesign across waveform structure, synchronization, spatial processing, and electromagnetic hardware.

In that sense, the next generation of ISABC systems will likely move beyond conventional passive tags toward more intelligent reflective platforms. MIMO backscatter, polarization diversity, beam-aware architectures, and metasurface- or lens-assisted reflectors can provide new degrees of freedom to improve robustness, coverage, and spectral efficiency. As 6G research continues to push ISAC into mainstream wireless system design, ISABC stands out as a particularly attractive solution for low-power IoT and smart infrastructure scenarios, where battery replacement, hardware complexity, and deployment cost must be minimized. Its long-term significance lies in showing that sensing and communication need not be added as separate functions but can instead be embedded together into the physics of reflection itself.

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